

Advanced Class Imbalance Handling: SMOTE vs. GANs

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Abstract

The handling of extreme class imbalance is examined using a binary MNIST task. Three data-level approaches are compared: SMOTE interpolation, geometric augmentation, and conditional GAN generation. Geometric augmentation improves feature learning but fails to produce usable minority detections under severe imbalance. SMOTE is best for minority classes with a simple geometric structure. For topologically complex digits, structural properties are better preserved by GAN-based generation.

Problem Setup & Methodology

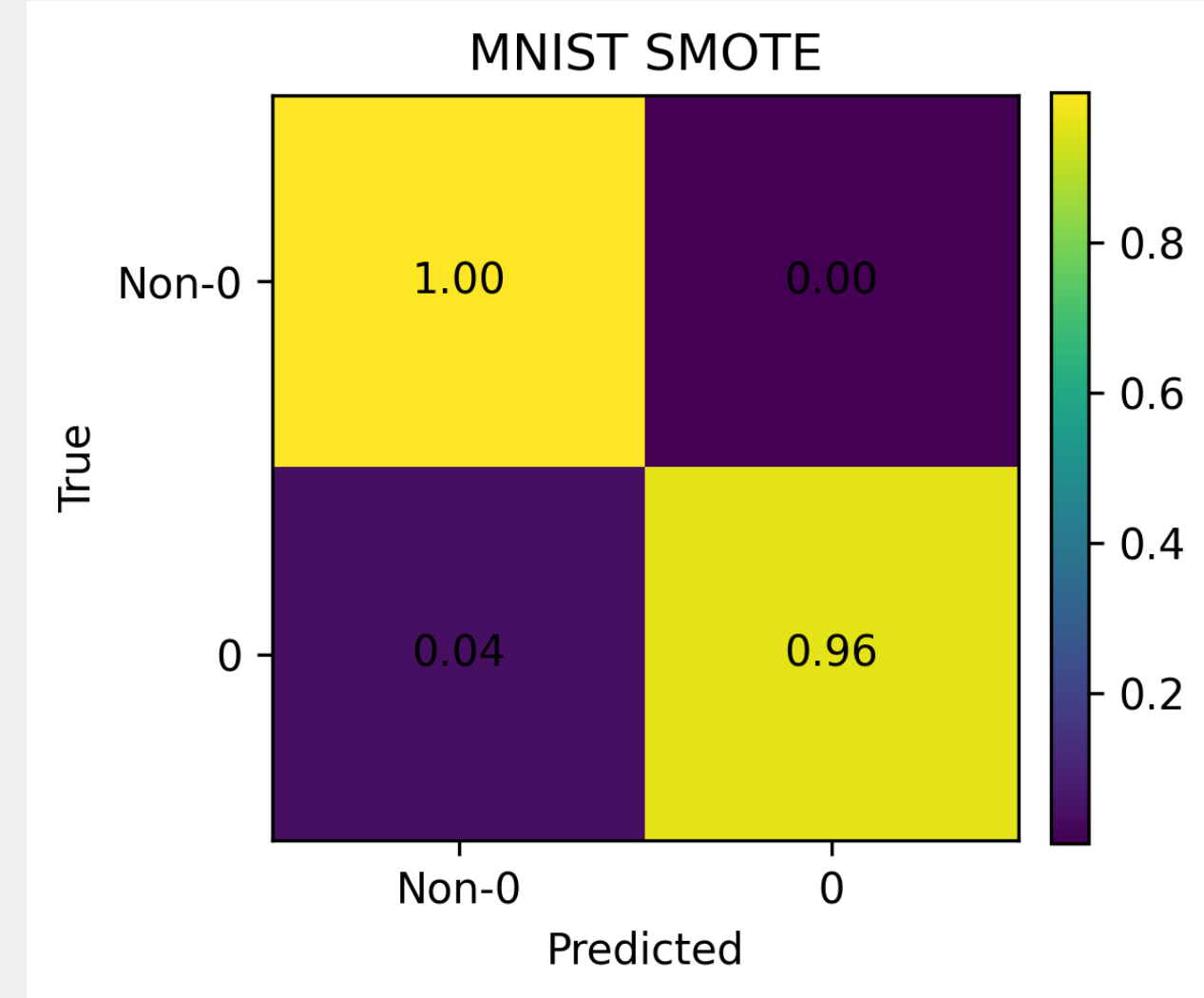
Dataset: A binary MNIST classification task (Digit 0 vs Rest) is considered, with a **1:50 class imbalance**.

Minority Samples: Approximately 120 images are available.

Objective: Minority detection quality is evaluated under extreme imbalance.

- SMOTE:** Synthetic samples are generated via linear interpolation between minority neighbors.
- Geometric Augmentation:** Input images are rotated ($\pm 15^\circ$) without increasing class count.
- Conditional GAN:** A deep generator is trained exclusively on minority-class samples.

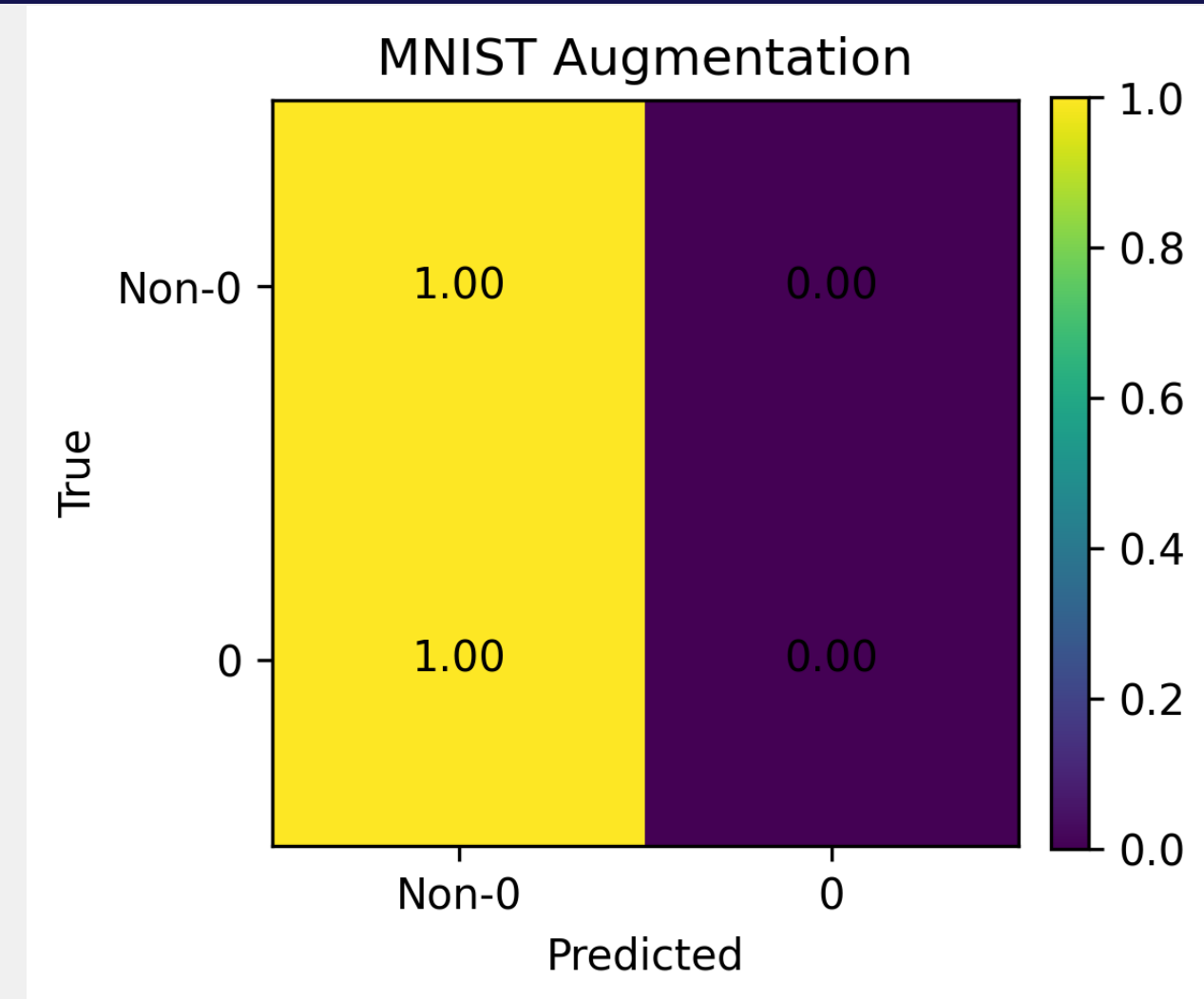
Experiment 1: Baseline vs. SMOTE



Result: Near-perfect separation is achieved (**0.96 Recall**).

Insight: Due to the simple topology of the digit, linear interpolation is sufficient for generating decision-consistent samples.

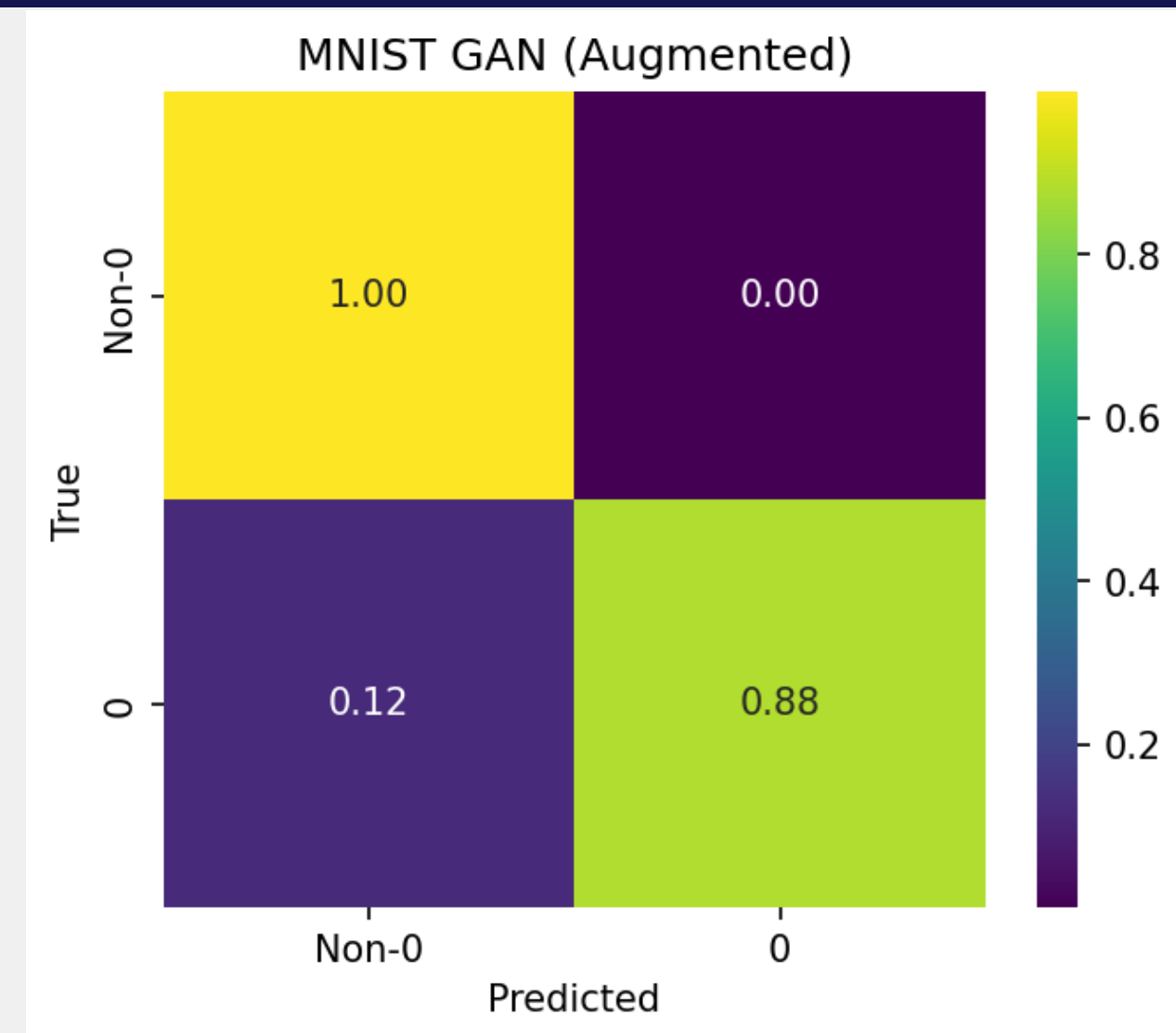
Experiment 2: The Augmentation Paradox



Result: High ranking performance is observed, while **zero precision** is obtained.

Explanation: At the selected decision threshold, predictions are assigned exclusively to the majority class.

Experiment 3: GAN Generation



Result: Strong recall (0.88), noisier boundary.

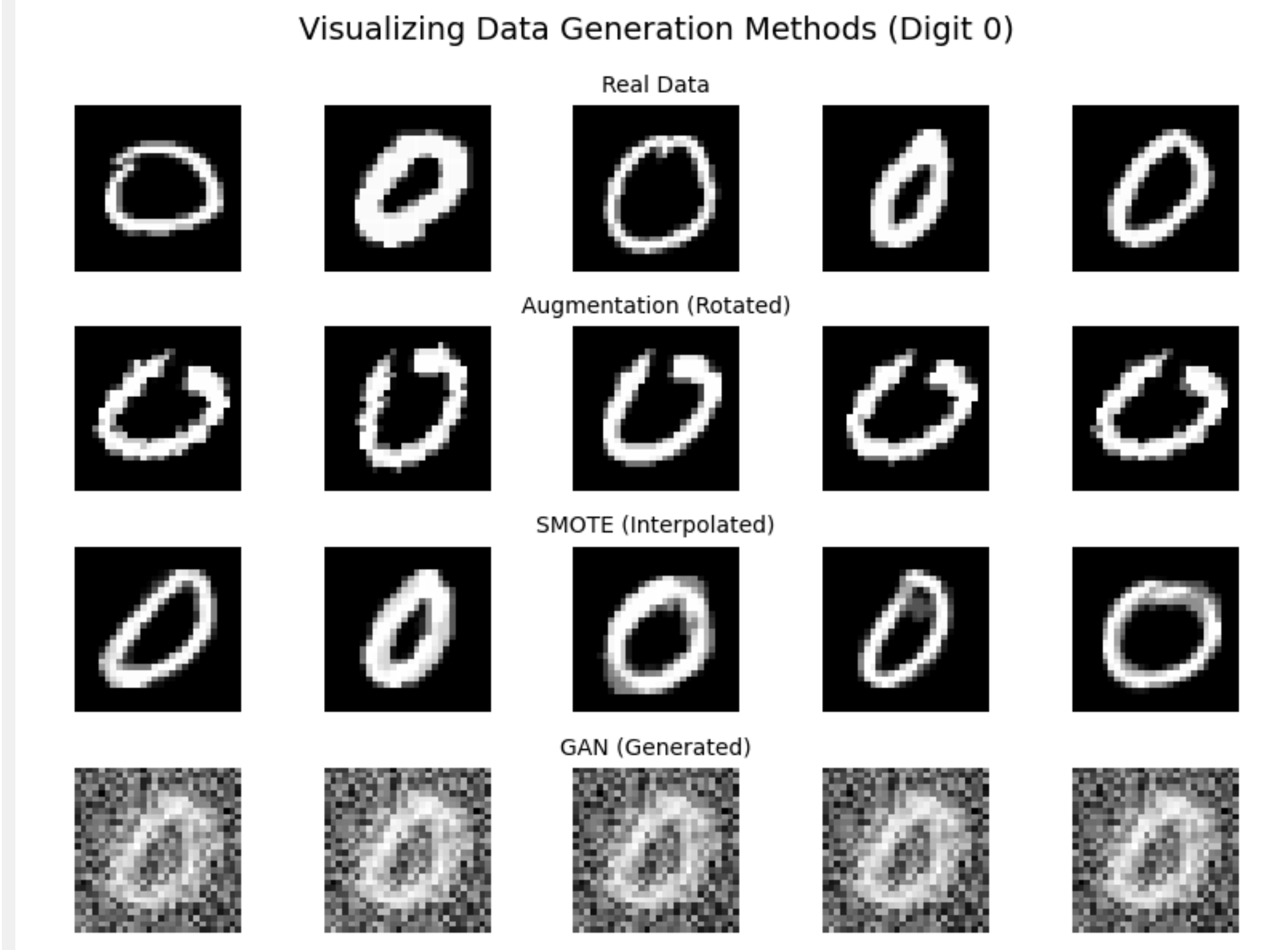
Challenge: Learning full distributions from ≈ 100 samples.

Why Precision Matters

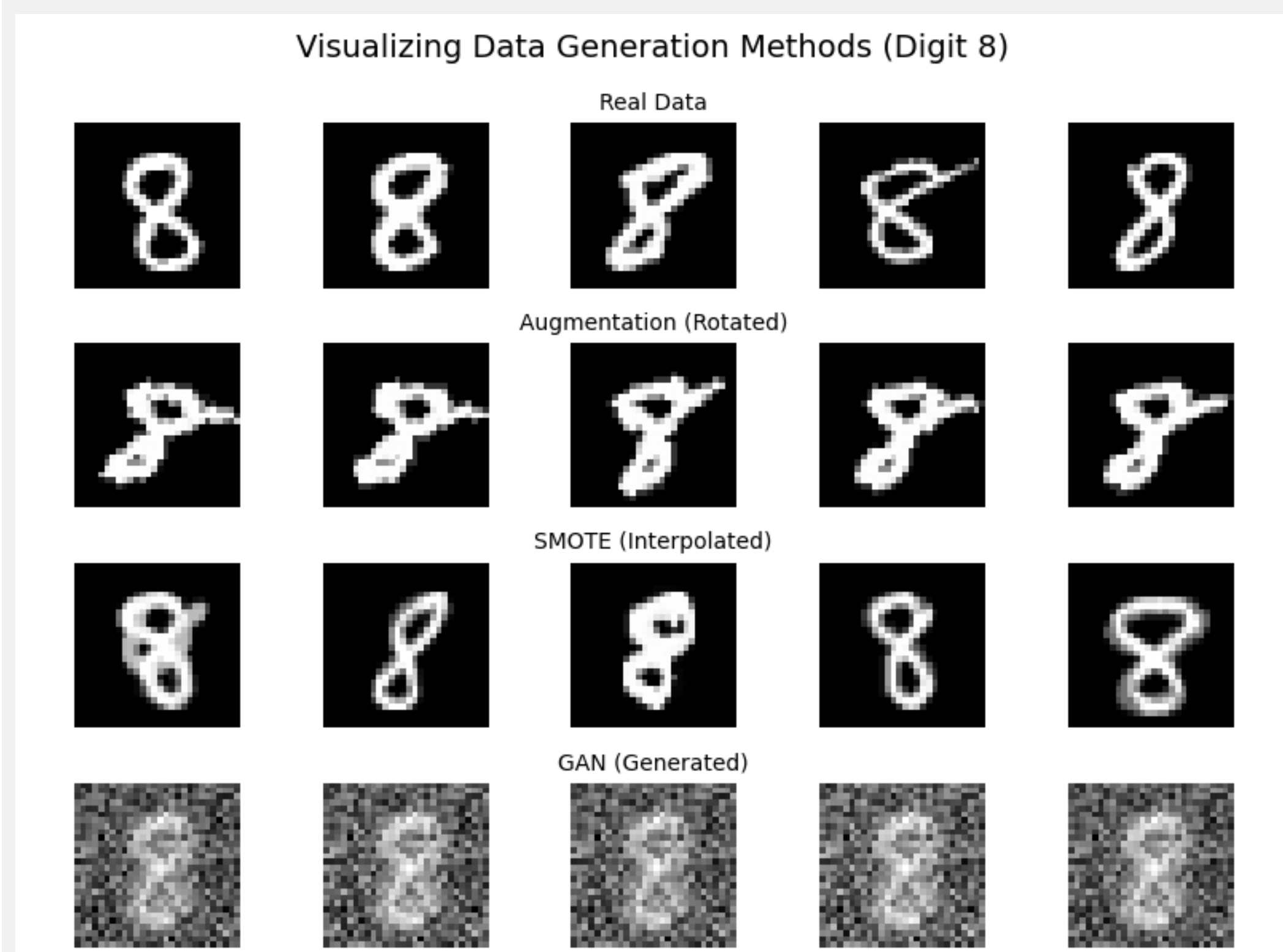
Under extreme imbalance, **accuracy and ROC-AUC are misleading**. Predicting the majority class yields high accuracy but **zero usable detections**. We therefore prioritize **PR-AUC**.

All confusion matrices are shown at threshold 0.5, reflecting deployment-time decisions.

Visual Comparison: Simple vs. Complex Topology



Digit 0: SMOTE preserves topology; GAN adds variance.



Digit 8: GAN preserves crossings; SMOTE introduces artifacts.

Quantitative Benchmarks (Decision Metrics)

Method	PR-AUC	ROC-AUC	Status
Baseline (Weights)	0.41	0.9953	Reference
SMOTE	0.96	0.9997	Best
Augmentation	0.00	0.9979	Fail
GAN (cGAN)	0.88	≈ 0.99	Competitive

Conclusion

- SMOTE is shown to be effective for minority classes with simple geometric structure.
- GAN-based generation is found to better preserve complex topological features.
- Geometric augmentation alone is insufficient for correcting extreme class imbalance.